

Selected Research Gap

While most robotic space missions have focused on a single agent carrying all the capabilities needed to complete the mission objectives, the future of advanced planetary exploration lies in multi-agent systems that are able to share responsibility and support each other. In order to fully understand a body as complex as Europa, it is necessary to divide surveying tasks across the main exploration rover and the orbital reconnaissance agents needed to scout terrain ahead of it. My mission will focus on designing a robust system of agents that can orbit Europa and perform reconnaissance for a rover traversing the surface in search of a proper drilling site to access the inner layers of the moon. This mission directly addresses three converging research gaps: autonomous operations in GPS-denied, high-radiation environments [1]; multi-agent coordination and interoperability for cooperative task planning [2]; and fully autonomous approach and proximity operations without Earth-based navigational support [3].

Mission Objective Statement:

A coordinated multi-agent system deployed to Europa's surface to autonomously characterize the ice shell and identify an optimal cryobot insertion site, operating without real-time Earth support.

Mission Destination

My mission would take place on Europa, Jupiter's icy moon. I chose this destination due to the abundance of knowledge there is to gain from an in-depth exploration of the system. Europa holds strong indicators of conditions favorable to life and contains all the chemical building blocks necessary to support it [1]. The layers of ice introduce in-situ resources that might benefit further exploration of the moon as long-term infrastructure is developed, as well as imposing significant rover design constraints. Any surface vehicle must feature traction-optimized wheels capable of accurately traversing the fractured, chaotic terrain.

Europa is also incredibly challenging. The 44-minute one-way light delay means every decision from landing onward must be fully autonomous [3]. Due to the intense radiation environment driven by Jupiter's magnetosphere, the rover would rely on orbiting satellite robots to scan the terrain and share terrain maps prior to surface traversal. Such a system would have to be highly robust to faulted or adversarial agents, as incorrect or incomplete maps shared across the swarm are likely to produce dangerous navigation decisions.

This location is essential to the mission because it provides an incredibly challenging environment that will necessitate smart, advanced robot design and algorithm development. It is the perfect stage to push robotic capabilities to their limits.

Environmental & Operational Challenges

Europa's environment is hostile in ways that shape every design decision in this mission. Jupiter's magnetosphere floods the moon's surface with high-energy charged particles, exposing electronics to radiation doses far greater than what commercial components can tolerate. This motivates a requirement for radiation-hardened electronics and a shielded vault setup for the rover's core computing hardware, similar to the approach taken on the Galileo spacecraft [1]. Redundancy at the agent level is an equally important mitigation. If one surface relay node builds-up enough single-event upsets to go silent, the remaining agents must autonomously reconfigure the network to maintain coverage.

Surface temperatures on Europa are around 90 K, which introduces thermal management challenges for both batteries and actuators. Power storage capacity drops significantly at cryogenic temperatures, meaning the rover must actively manage its thermal systems to keep cells within an operational range rather than simply relying on passive insulation.

The complete absence of GPS and the communication blackout due to the 44-minute light delay means the swarm cannot depend on any external reference for localization or replanning. Each agent must maintain its own state estimate using onboard sensors and inter-agent ranging, and the swarm must reach consensus on shared map data using algorithms that are resilient to packet loss and radiation-induced faults [2, 3]. There is no human in the loop to catch a bad map before the rover commits to a traverse.

Required Robot Capabilities

Mobility:

The rover must be able to navigate Europa's icy, fractured terrain and accurately drive without losing localization. It should be capable of traversing mild inclines and obstacles commonly encountered in planetary surface missions, with wheel geometry selected for the low-gravity, low-traction conditions specific to Europa's surface.

Autonomy:

The robot shall localize itself using onboard sensor data and inter-agent range measurements in the absence of GPS, generate a planned traverse path, and accurately execute that path. It should also analyze sensor readings to prioritize target areas based on mission objectives. Ground-penetrating radar sounder data indicating ice shell thickness will directly inform site selection logic, guiding the rover toward the thinnest viable ice for cryobot deployment.

Power Systems:

Due to the rover's distance from the sun, solar power is not viable. Instead, the system must rely on a Radioisotope Thermoelectric Generator (RTG) or similar nuclear power source. Power budget management is a crucial. The robot should track consumption across planned traverses and coordinate with the swarm to avoid scheduling high power draw operations simultaneously across multiple agents.

Communication:

The robot must communicate with surface relay nodes to exchange sensor readings, map updates, and mission state. This local mesh network is the primary coordination channel for the swarm. Communication with Earth still exists as a secondary channel but is non-deterministic due to the 44-minute delay and periodic occultation by Jupiter. The inter-agent protocol must tolerate partial dropouts and continue operating under degraded connectivity without stalling consensus.

Durability & Specialized Instruments:

The rover shall be designed to withstand Europa's radiation and cryogenic environment, targeting operation across the full mission lifecycle. It shall carry a ground-penetrating radar sounder (analogous to the REASON instrument used for Europa Clipper) to characterize ice shell thickness and build a subsurface map as it traverses the surface in search of the optimal cryobot insertion point [1].

References

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- [3] NASA, “FY 2026–2027 SBIR/STTR Program Solicitation, Appendix 26B-I: SBIR Topics,” NASA Small Business Innovation Research Program, 2025.